In [39]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
import statsmodels.formula.api as smf
```

In [40]:
```
one_job_per_thread = pd.read_csv('./integral_parallel_measurements/measur
one_job_per_thread = one_job_per_thread.drop(one_job_per_thread.index[-1]
three_jobs_per_thread = pd.read_csv('./integral_parallel_measurements/mea
three_jobs_per_thread = three_jobs_per_thread.drop(three_jobs_per_thread.
eight_jobs_per_thread = pd.read_csv('./integral_parallel_measurements/mea
eight_jobs_per_thread = eight_jobs_per_thread.drop(eight_jobs_per_thread.

one_job_per_thread['partitions'] = 1
three_jobs_per_thread['partitions'] = 3
eight_jobs_per_thread['partitions'] = 8

df = pd.concat([one_job_per_thread, three_jobs_per_thread, eight_jobs_per
df['threads_num'] = df["threads/func"]
df = df.drop("threads/func", axis=1)
```

Function 1

We have decided to throw out measurements for 100 and 1000 number of threads as they are outliers.

In [41]:
```
pd.concat([one_job_per_thread.head(2), three_jobs_per_thread.tail(2), eig
```

Out[41]:

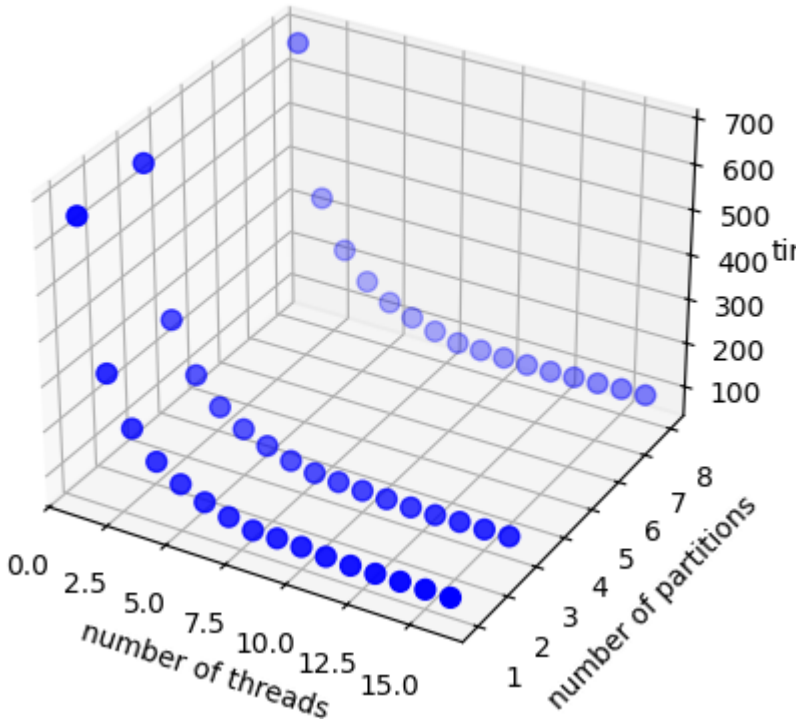
	threads/func	1	2	3	partitions
0	1	658.2	789.0	5.0	1
1	2	334.0	403.0	3.0	1
14	15	80.2	95.8	6.8	3
15	16	81.8	97.8	7.6	3
0	1	674.4	820.0	5.2	8
1	2	336.2	415.0	3.0	8

In [42]:
```
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

ax.scatter(df["threads_num"], df["partitions"], df["1"], c='blue', marker

ax.set_xlabel("number of threads")
ax.set_ylabel("number of partitions")
ax.set_zlabel("time")

plt.show()
```



In [43]:
```
exec_time_model = smf.ols(formula='1 ~ threads_num + partitions', data=df
exec_time_model.summary()
```

```
/Users/vitya/Documents/acs/lab5-threadpool-integral-valihursyi_shtohryn_ta
rnavskyi/.env/lib/python3.11/site-packages/statsmodels/regression/linear_m
odel.py:1782: RuntimeWarning: divide by zero encountered in scalar divide
  return 1 - self.ssr/self.centered_tss
```

Out[43]:

OLS Regression Results

Dep. Variable:	Intercept	**R-squared:**	-inf
Model:	OLS	**Adj. R-squared:**	-inf
Method:	Least Squares	**F-statistic:**	-22.50
Date:	Wed, 19 Mar 2025	**Prob (F-statistic):**	1.00
Time:	23:05:34	**Log-Likelihood:**	1573.3
No. Observations:	48	**AIC:**	-3141.
Df Residuals:	45	**BIC:**	-3135.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	1.0000	5.25e-16	1.91e+15	0.000	1.000	1.000
threads_num	6.592e-17	4.56e-17	1.447	0.155	-2.58e-17	1.58e-16
partitions	-2.602e-17	7.13e-17	-0.365	0.717	-1.7e-16	1.18e-16

Omnibus:	3.882	**Durbin-Watson:**	0.039
Prob(Omnibus):	0.144	**Jarque-Bera (JB):**	1.767
Skew:	0.022	**Prob(JB):**	0.413
Kurtosis:	2.061	**Cond. No.**	26.1

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

P-value of the coefficient near number of partitions is very high, thus we consider it statistically insignificant for the following estimations. P-value of the t-test for coefficient near threads_num is also high, but setting size of test high enough we can accept null hypothesis.

Let $speedup_{obs}(n) = \frac{\text{execution on 1 thread}}{\text{execution on } n \text{ threads}}$ be observable speedup. Amdal's law states that $speedup(n) = \frac{1}{(1-\alpha)+\frac{\alpha}{n}}$, where α is the part of the task that can be parallelized. Slightly rewriting equation:

$$\frac{1}{speedup(n)} = (1-\alpha) + \frac{\alpha}{n}$$
$$\frac{1}{speedup(n)} - 1 = \alpha(\frac{1}{n} - 1)$$

We set $speedup(n)$ to be $speedup_{obs}(n)$. Now, we can estimate the parameter using OLS simple linear regression.

In [44]:
```
df = one_job_per_thread

baseline = df[(df["threads/func"] == 1) & (df["partitions"] == 1)]["1"].v
df["speedup_obs"] = baseline/df["1"]

df["predicted"] = 1/df["speedup_obs"]-1
df["predictor"] = 1/df["threads/func"]-1

speedup_model = smf.ols(formula='predicted ~ predictor-1', data=df).fit()


print(f"alpha estimated to be:", round(speedup_model.params["predictor"],

print(np.sum(df["predicted"]/df["predictor"])/len(df["predicted"]))
```

```
alpha estimated to be: 0.971
0.9124606835608431
```

We get an outstanding result of 97% paralellized looking at coefficient. However as we do not have intercept in this model, it makes it biased. Thus, this result should not be taken too seriously, as we used linear regression, and the relation we were looking for could have been not really linear in parameters. Also, estimating parameters by coefficients is bad practice, as coefficients depend on the scale of the variable.

I have tried more mathematically appropriate method

$$\alpha = \frac{\frac{1}{speedup(n)} - 1}{\frac{1}{n} - 1}$$

Took average of obtained value and got result of 0.91 which is more down to earth.

In [45]:
```
speedup_model.summary()
```

```
/Users/vitya/Documents/acs/lab5-threadpool-integral-valihursyi_shtohryn_ta
rnavskyi/.env/lib/python3.11/site-packages/scipy/stats/_axis_nan_policy.p
y:430: UserWarning: `kurtosistest` p-value may be inaccurate with fewer th
an 20 observations; only n=16 observations were given.
  return hypotest_fun_in(*args, **kwds)
```

Out[45]:

OLS Regression Results

Dep. Variable:	predicted	**R-squared (uncentered):**	1.000
Model:	OLS	**Adj. R-squared (uncentered):**	1.000
Method:	Least Squares	**F-statistic:**	4.499e+04
Date:	Wed, 19 Mar 2025	**Prob (F-statistic):**	1.69e-27
Time:	23:05:34	**Log-Likelihood:**	44.946
No. Observations:	16	**AIC:**	-87.89
Df Residuals:	15	**BIC:**	-87.12
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
predictor	0.9709	0.005	212.107	0.000	0.961	0.981

Omnibus:	1.305	**Durbin-Watson:**	0.143
Prob(Omnibus):	0.521	**Jarque-Bera (JB):**	1.072
Skew:	0.454	**Prob(JB):**	0.585
Kurtosis:	2.115	**Cond. No.**	1.00

Notes:
[1] R^2 is computed without centering (uncentered) since the model does not contain a constant.
[2] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Even though, p-value for the α is very small, so it is statistically significant. R-squared is 1, which means that model explains 100% of variance of data.